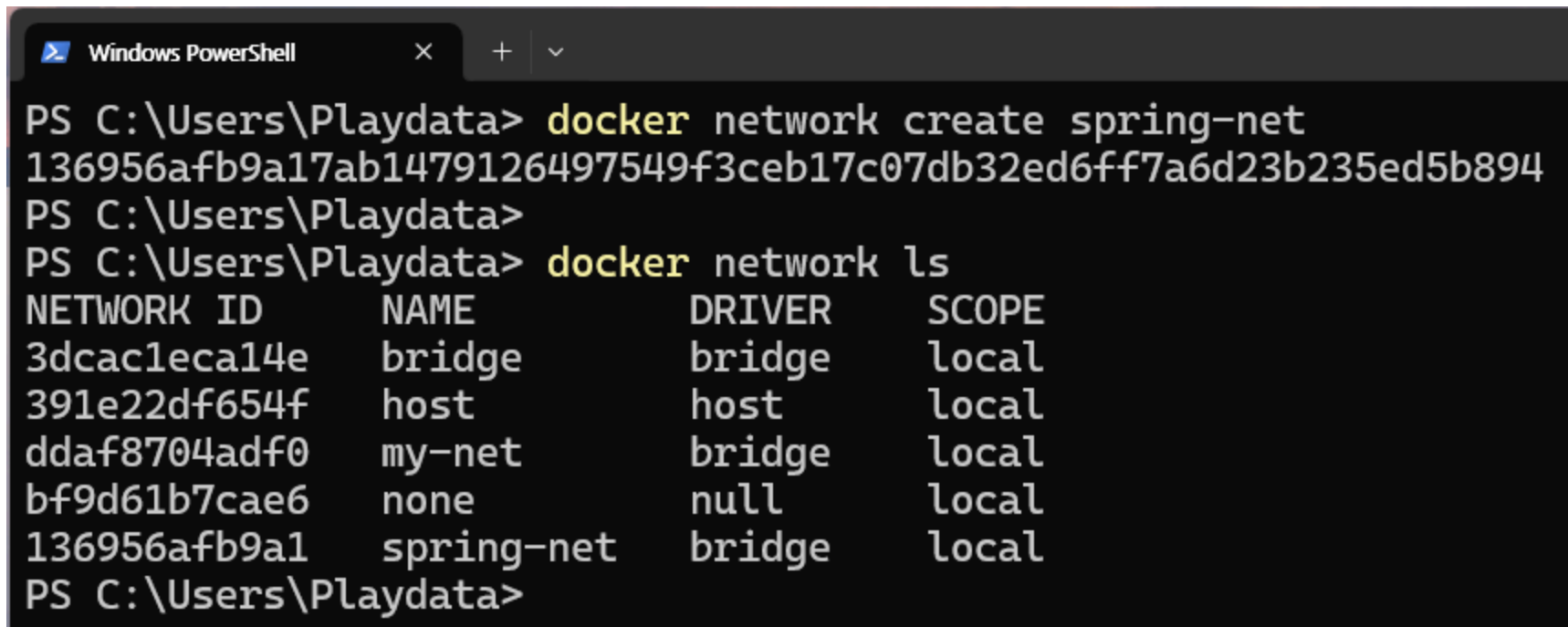


Docker Network

Docker Network 생성

```
docker network create spring-net  
docker network ls
```



```
Windows PowerShell
PS C:\Users\Playdata> docker network create spring-net
136956afb9a17ab1479126497549f3ceb17c07db32ed6ff7a6d23b235ed5b894
PS C:\Users\Playdata>
PS C:\Users\Playdata> docker network ls
```

NETWORK ID	NAME	DRIVER	SCOPE
3dcac1eca14e	bridge	bridge	local
391e22df654f	host	host	local
ddaf8704adf0	my-net	bridge	local
bf9d61b7cae6	none	null	local
136956afb9a1	spring-net	bridge	local

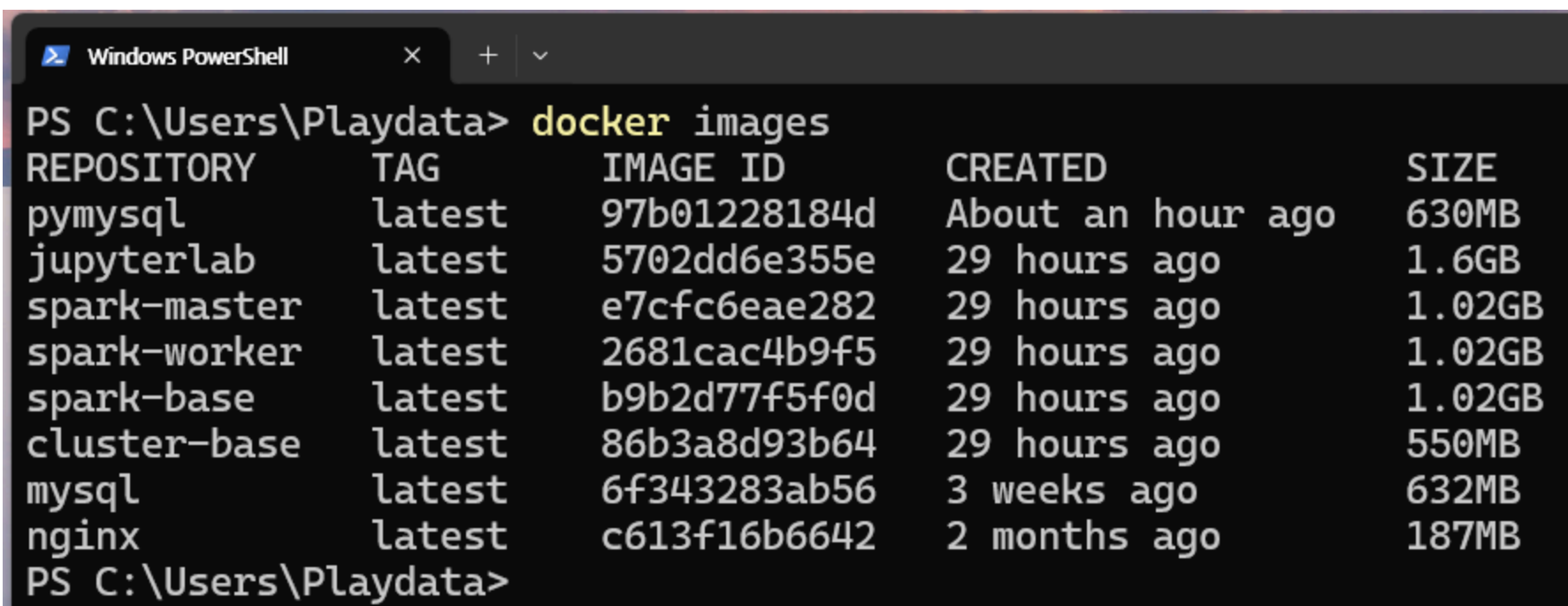
```
PS C:\Users\Playdata>
```

MySQL

단계1: MySQL Image

```
docker images
```

```
# 만약 없다면,  
docker pull mysql
```



```
Windows PowerShell
PS C:\Users\Playdata> docker images
```

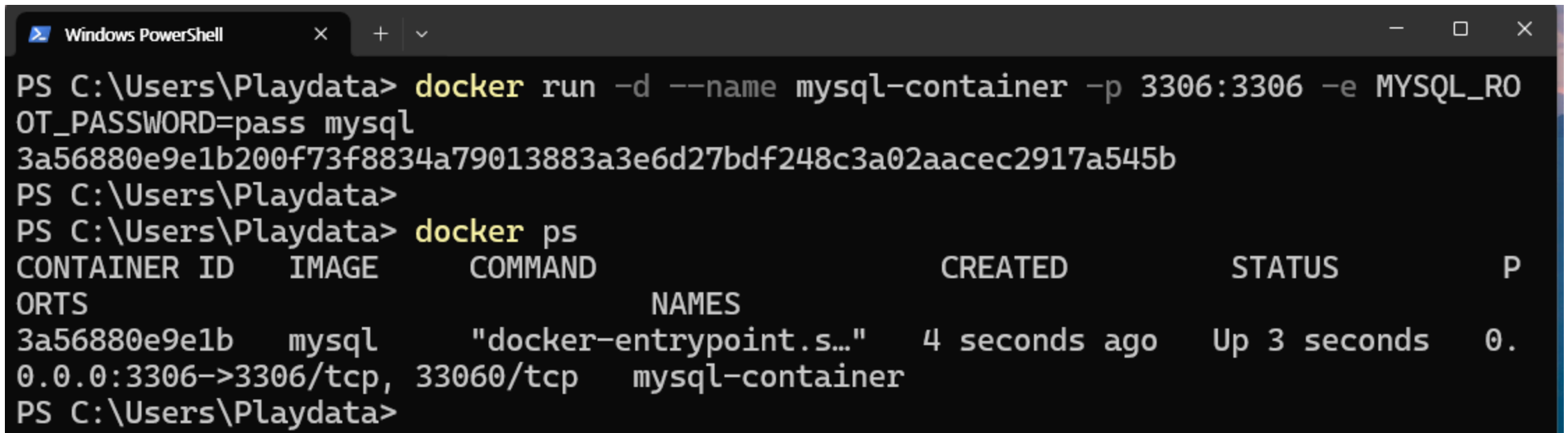
REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
pymysql	latest	97b01228184d	About an hour ago	630MB
jupyterlab	latest	5702dd6e355e	29 hours ago	1.6GB
spark-master	latest	e7cfc6eae282	29 hours ago	1.02GB
spark-worker	latest	2681cac4b9f5	29 hours ago	1.02GB
spark-base	latest	b9b2d77f5f0d	29 hours ago	1.02GB
cluster-base	latest	86b3a8d93b64	29 hours ago	550MB
mysql	latest	6f343283ab56	3 weeks ago	632MB
nginx	latest	c613f16b6642	2 months ago	187MB

```
PS C:\Users\Playdata>
```

단계2: MySQL Container

```
docker run -d --name mysql-container -p 3306:3306 -e MYSQL_ROOT_PASSWORD=pass mysql

docker ps
```

A screenshot of a Windows PowerShell terminal window. The title bar shows 'Windows PowerShell' with standard window controls. The terminal content shows the execution of Docker commands. First, 'docker run -d --name mysql-container -p 3306:3306 -e MYSQL_ROOT_PASSWORD=pass mysql' is executed, resulting in a long container ID. Then, 'docker ps' is executed, displaying a table of running containers. The table has columns for Container ID, Image, Command, Names, Created, Status, and Ports. One container is listed: '3a56880e9e1b' with image 'mysql', command '"docker-entrypoint.s...", and name 'mysql-container'. It was created '4 seconds ago' and is 'Up 3 seconds'.

```
PS C:\Users\Playdata> docker run -d --name mysql-container -p 3306:3306 -e MYSQL_ROOT_PASSWORD=pass mysql
3a56880e9e1b200f73f8834a79013883a3e6d27bdf248c3a02aacec2917a545b
PS C:\Users\Playdata>
PS C:\Users\Playdata> docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS
3a56880e9e1b   mysql    "docker-entrypoint.s...  4 seconds ago Up 3 seconds  0.0.0:3306->3306/tcp, 33060/tcp
PS C:\Users\Playdata>
```

단계3: dbeaver 및 CREATE DATABASE

```
CREATE DATABASE springdocker;
```

```
show databases;
```

The screenshot shows the DBeaver IDE interface. The top panel displays a SQL script with two commands: `CREATE DATABASE springdocker;` and `show databases;`. The bottom panel shows the results of the `show databases;` command, listing the databases: `information_schema`, `mysql`, `performance_schema`, `springdocker`, and `sys`.

Database Navigator: 프로젝트

Enter a part of object name here

root(spring)@MySQL - localhost:3306

- Databases
 - sys
 - Users
 - Administer
 - System Info
- root@MySQL - localhost:3306
- urstory@MySQL - localhost:3306

*root(spring)@MySQL> Script-3

```
CREATE DATABASE springdocker;
```

```
show databases;
```

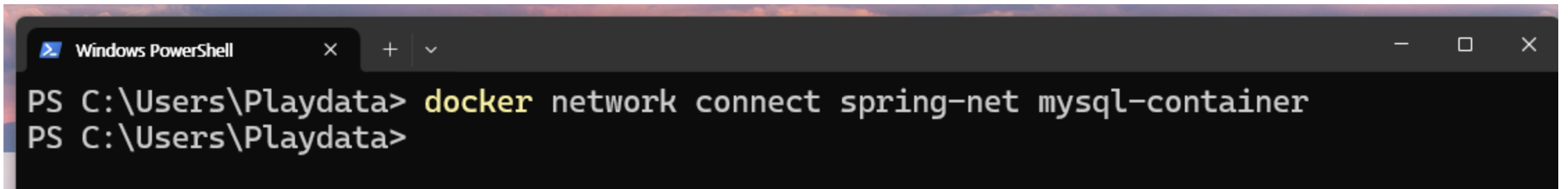
schemata 1

show databases | Enter a SQL expression to filter results (use Ctrl+Space)

	asc Database	
1	information_schema	
2	mysql	
3	performance_schema	
4	springdocker	
5	sys	

단계4: Network Connect

```
docker network connect spring-net mysql-container
```



```
Windows PowerShell
PS C:\Users\Playdata> docker network connect spring-net mysql-container
PS C:\Users\Playdata>
```

SpringBoot JPA

단계1: dependencies

```
dependencies {  
    implementation 'org.springframework.boot:spring-boot-starter-data-jpa'  
    implementation 'org.springframework.boot:spring-boot-starter-web'  
    compileOnly 'org.projectlombok:lombok'  
    runtimeOnly 'com.mysql:mysql-connector-j'  
    annotationProcessor 'org.projectlombok:lombok'  
    testImplementation 'org.springframework.boot:spring-boot-starter-test'  
}
```

단계2: application.yaml

```
spring:
  output:
    ansi:
      enabled: always
  # 데이터베이스 설정
  datasource:
    # jdbc:mysql://[mysql 도커 컨테이너명]:3306/[database이름]
    url: jdbc:mysql://mysql-container:3306/springdocker?userSSL=false&allowPublicKeyRetrieval=true&serverTimezone=Asia/Seoul
    username: root
    password: pass
    driver-class-name: com.mysql.cj.jdbc.Driver
  # jpa 설정
  jpa:
    hibernate:
      ddl-auto: update
    properties:
      hibernate:
        show_sql: true
        format_sql: true
```

단계3: @SpringBootTest & @Test 주석처리

- Test 과정에서 JPA가 MySQL 접속하는 과정에 오류가 발생할 수 있음
- 아직 MySQL이 포함된 docker network에 접속을 할 수 없기 때문에 오류가 발생함

```
EXPLORER
...
SpringdockernetworkApplicationTests.java 2 x

src > test > java > com > example > springdockernetwork > Springdockern

1  package com.example.springdockernetwork;
2
3  import org.junit.jupiter.api.Test;
4  import org.springframework.boot.test.context.SpringBootTest;
5
6  // @SpringBootTest
7  class SpringdockernetworkApplicationTests {
8
9      // @Test
10     void contextLoads() {
11     }
12
13 }
```

단계4: Dockerfile

```
FROM gradle:7.6-jdk17-alpine as build

ENV APP_HOME=/apps

WORKDIR $APP_HOME

COPY build.gradle settings.gradle gradlew $APP_HOME

COPY gradle $APP_HOME/gradle

RUN chmod +x gradlew

RUN ./gradlew build || return 0

COPY src $APP_HOME/src

RUN ./gradlew clean build

FROM openjdk:17.0.2-jdk

ENV APP_HOME=/apps
ARG ARTIFACT_NAME=app.jar
# ARG JAR_FILE_PATH=build/libs/[프로젝트명]-0.0.1-SNAPSHOT.jar
ARG JAR_FILE_PATH=build/libs/springdockernetwork-0.0.1-SNAPSHOT.jar

WORKDIR $APP_HOME

COPY --from=build $APP_HOME/$JAR_FILE_PATH $ARTIFACT_NAME

EXPOSE 8080

#ENTRYPOINT ["java", "-XX:+UnlockExperimentalVMOptions", "-XX:+UseCGroupMemoryLimitForHeap", "-Djava.security.egd=file:/dev/./urandom", "-jar", "app.jar"]
ENTRYPOINT ["java", "-jar", "app.jar"]
```

단계5: docker build

```
docker build -t springdockernetwork .
```

The screenshot shows an IDE interface with a file explorer on the left and a terminal on the right. The file explorer displays a project named 'SPRING...' with files: '.gradle', '.vscode', 'build', 'gradle', 'src' (highlighted), '.gitignore', 'build.gradle', and 'Dockerfile'. The terminal shows a command prompt at 'PS C:\dev\docker\docker_network\springdockernetwork>' where the command 'docker build -t springdockernetwork .' has been executed. The output shows the build progress: '[+] Building 13.0s (14/18)' and various internal steps like 'load build definition from Dockerfile', 'transferring dockerfile: 833B', 'load metadata for docker.io/library/openjdk:17.0.2-jdk', 'load metadata for docker.io/library/gradle:7.6-jdk17-alpine', 'library/openjdk:pull token for registry-1.docker.io', 'library/gradle:pull token for registry-1.docker.io', 'load .dockerignore', 'transferring context: 2B', and 'FROM docker.io/library/openjdk:17.0.2-jdk@sha256:528707081fdb9562eb819128'. The terminal output is color-coded with green for success and yellow for warnings.

단계6: springdockernetwork

docker images

```

SPRINGDOCKERNETWORK
├── gradle
├── src
│   ├── main
│   │   ├── java\com\exa...
│   │   │   ├── model
│   │   │   │   ├── DockerEntit...
│   │   │   │   ├── DockerRepo...
│   │   │   │   └── Springdocker...
│   │   └── resources
│   │       └── static
└── ...

PS C:\dev\docker\docker_network\springdockernetwork> docker images
REPOSITORY              TAG               IMAGE ID           CREATED            SIZE
springdockernetwork     latest            3782f0ab8e8a       11 seconds ago    518MB
jupyterlab               latest            5702dd6e355e       2 days ago        1.6GB
spark-base               latest            b9b2d77f5f0d       2 days ago        1.02GB
spark-master             latest            e7cfc6eae282       2 days ago        1.02GB
spark-worker             latest            2681cac4b9f5       2 days ago        1.02GB
cluster-base             latest            86b3a8d93b64       2 days ago        550MB
mysql                    latest            6f343283ab56       3 weeks ago       632MB
nginx                    latest            c613f16b6642       2 months ago      187MB
mysql                    5.7              5107333e08a8       4 months ago      501MB

PS C:\dev\docker\docker_network\springdockernetwork>

```

단계7: docker run

```
docker run -d --name spring-container --network=spring-net springdockernetwork
```

```
docker ps
```

```
PS C:\dev\docker\docker_network\springdockernetwork> docker run -d --name spring-container --network=spring-net springdockernetwork
6e529848b2e7c534f030c829fe9567c176a156d5333e7b425895fa8adb114c15
● PS C:\dev\docker\docker_network\springdockernetwork> docker ps
```

CONTAINER ID	IMAGE	NAMES	COMMAND	CREATED	STATUS	PORTS
6e529848b2e7	springdockernetwork	spring-container	"java -jar app.jar"	10 seconds ago	Up 9 seconds	8080/tcp
daa2533d184b	mysql	mysql-container	"docker-entrypoint.s..."	8 minutes ago	Up 8 minutes	0.0.0.0:3306->3306/tcp, 33060/tcp
403adf326f6e	spark-worker	spark-worker-1	"/bin/sh -c 'bin/spa..."	31 minutes ago	Up 31 minutes	0.0.0.0:8081->8081/tcp
1c46a1399f3d	spark-worker	spark-worker-2	"/bin/sh -c 'bin/spa..."	31 minutes ago	Up 31 minutes	0.0.0.0:8082->8081/tcp
ba5b542837b3	spark-master	spark-master	"/bin/sh -c 'bin/spa..."	31 minutes ago	Up 31 minutes	0.0.0.0:7077->7077/tcp, 0.0.0.0:8080->8080/tcp
7cb67f58b421	jupyterlab	jupyterlab	"/bin/sh -c 'jupyter..."	31 minutes ago	Up 31 minutes	0.0.0.0:8888->8888/tcp

```
○ PS C:\dev\docker\docker_network\springdockernetwork> █
```

단계8: springboot jpa와 mysql 연동확인

```
use springdocker;
```

```
show tables;
```

